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Andrew Thon Riem Alier.

Iteration(1)

Assumptions:
- Learning rate $\alpha = 0.01$ (not given, so we chose it)
- b is a 2x1 vector meaning one bias per data point.

Setup.

$$X = \begin{pmatrix} 1 & 3 \\ 4 & 10 \end{pmatrix}, \quad y = \begin{pmatrix} 5 \\ 6 \end{pmatrix}, \quad m^{(0)} \begin{pmatrix} -1 \\ 2 \end{pmatrix}, \quad b^{(0)} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad n = 2$$

Model and cost:

$$\hat{y} = xm + b, \quad e = \hat{y} - y, \quad J = \frac{1}{n} e^T e$$

Derivation of gradients (chain rule)

Soln.

$$\frac{\partial J}{\partial \hat{y}} = \frac{2}{n}(\hat{y} - y) = \frac{2}{n} e$$

$$\frac{\partial \hat{y}}{\partial m} = X, \quad \frac{\partial \hat{y}}{\partial b} = I$$

Chain rule:

$$\frac{\partial J}{\partial m} = \left(\frac{\partial \hat{y}}{\partial m}\right)^T \frac{\partial J}{\partial \hat{y}} = X^T \cdot \frac{2}{n} e = \frac{2}{n} X^T (\hat{y} - y)$$

$$\frac{\partial J}{\partial b} = \left(\frac{\partial \hat{y}}{\partial b}\right)^T \frac{\partial J}{\partial \hat{y}} = I^T \cdot \frac{2}{n} e = \frac{2}{n} (\hat{y} - y)$$

Soln.

$$m \leftarrow m - \alpha \frac{\partial J}{\partial m}, \quad b \leftarrow b - \alpha \frac{\partial J}{\partial b}.$$

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Iteration (1)

predictions:

$$X_m^{(0)} = \begin{bmatrix} 1(-1) + 3(2) \\ 4(-1) + 10(2) \end{bmatrix} = \begin{pmatrix} 5 \\ 16 \end{pmatrix}$$

$$\hat{y} = X_m^{(0)} + b^{(0)} = \begin{pmatrix} 5 \\ 16 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 6 \\ 17 \end{pmatrix}$$

Error Vector

$$e = \hat{y} - y = \begin{pmatrix} 6-5 \\ 17-6 \end{pmatrix} = \begin{pmatrix} 1 \\ 11 \end{pmatrix}$$

Cost

$$J = \frac{1}{n} e^T e = \frac{1^2 + 11^2}{2} = \frac{122}{2} = 61$$

Gradients

$$\frac{\partial J}{\partial m} = \frac{2}{n} X^T e = \begin{pmatrix} 1 & 4 \\ 3 & 10 \end{pmatrix} \begin{pmatrix} 1 \\ 11 \end{pmatrix} = \begin{pmatrix} 1(1) + 4(11) \\ 3(1) \quad 10(11) \end{pmatrix} = \begin{pmatrix} 45 \\ 113 \end{pmatrix}$$

$$\frac{\partial J}{\partial b} = \frac{2}{n} e = \begin{pmatrix} 1 \\ 11 \end{pmatrix}$$

Updates

$$m^{(1)} = m^{(0)} - \alpha \frac{\partial J}{\partial m} = \begin{pmatrix} -1 \\ 2 \end{pmatrix} - 0.01 \begin{pmatrix} 45 \\ 113 \end{pmatrix} = \begin{pmatrix} -1.45 \\ 0.87 \end{pmatrix}$$

$$b^{(1)} = b^{(0)} - \alpha \frac{\partial J}{\partial b} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} - 0.01 \begin{pmatrix} 1 \\ 11 \end{pmatrix} = \begin{pmatrix} 0.99 \\ 0.89 \end{pmatrix}$$

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iteration (2)

Using updated parameters from iteration(1)

- $m^{(1)} = \begin{pmatrix} -1.45 \\ 0.87 \end{pmatrix}$
- $b^{(1)} = \begin{pmatrix} 0.99 \\ 0.89 \end{pmatrix}$

1. Predictions ($\hat{y}$)

$$y = Xm^{(1)} + b^{(1)}$$

$$Xm^{(1)} = \begin{pmatrix} 1 & 3 \\ 4 & 10 \end{pmatrix}\begin{pmatrix} -1.45 \\ 0.87 \end{pmatrix} = \begin{pmatrix} 1(-1.45)+3(0.87) \\ 4(-1.45)+10(0.87) \end{pmatrix} = \begin{pmatrix} 1.16 \\ 2.90 \end{pmatrix}$$

$$\hat{y} = \begin{pmatrix} 1.16 \\ 2.90 \end{pmatrix} + \begin{pmatrix} 0.99 \\ 0.89 \end{pmatrix} = \begin{pmatrix} 2.15 \\ 3.79 \end{pmatrix}$$

2. Error Vector(e)

$$e = \hat{y} - y = \begin{pmatrix} 2.15-5 \\ 3.79-6 \end{pmatrix} = \begin{pmatrix} -2.85 \\ -2.21 \end{pmatrix}$$

3. Cost (J)

$$J = \frac{1}{n} e^T e = \frac{(-2.85)^2 + (-2.21)^2}{2} = \frac{8.1225 + 4.8841}{2} = \frac{13.0066}{2} = 6.5033$$

4. Gradients

$$\frac{\partial J}{\partial m} = \frac{2}{n} X^T e = \frac{2}{2}\begin{pmatrix} 1 & 4 \\ 3 & 10 \end{pmatrix}\begin{pmatrix} -2.85 \\ -2.21 \end{pmatrix} = \begin{pmatrix} 1(-2.85)+4(-2.21) \\ 3(-2.85)+10(-2.21) \end{pmatrix} = \begin{pmatrix} -11.69 \\ -30.65 \end{pmatrix}$$

$$\frac{\partial J}{\partial b} = \frac{2}{n} e = \frac{2}{2}\begin{pmatrix} -2.85 \\ -2.21 \end{pmatrix} = \begin{pmatrix} -2.85 \\ -2.21 \end{pmatrix}$$

5. Updates

$$m^{(2)} = m^{(1)} - \alpha \frac{\partial J}{\partial m} = \begin{pmatrix} -1.45 \\ 0.87 \end{pmatrix} - 0.01\begin{pmatrix} -11.69 \\ -30.65 \end{pmatrix} = \begin{pmatrix} -1.45+0.1169 \\ 0.87+0.3065 \end{pmatrix} = \begin{pmatrix} -1.3331 \\ 1.1765 \end{pmatrix}$$

$$b^{(2)} = b^{(1)} - \alpha \frac{\partial J}{\partial b} = \begin{pmatrix} 0.99 \\ 0.89 \end{pmatrix} - 0.01\begin{pmatrix} -2.85 \\ -2.21 \end{pmatrix} = \begin{pmatrix} 0.99+0.0285 \\ 0.89+0.0221 \end{pmatrix} = \begin{pmatrix} 1.0185 \\ 0.9121 \end{pmatrix}$$

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Iteration (3)

Using Updated parameters from Iteration (2)

. $m^{(2)} = \begin{pmatrix} -1.3331 \\ 1.1760 \end{pmatrix}$. $b^{(2)} = \begin{pmatrix} 1.0180 \\ 0.9121 \end{pmatrix}$

1. predictions: (ŷ) $\hat{y} = Xm^{(2)} + b^{(2)}$

$$Xm^{(2)} = \begin{pmatrix} 1 & 3 \\ 4 & 10 \end{pmatrix} \begin{pmatrix} -1.3331 \\ 1.1760 \end{pmatrix} = \begin{pmatrix} 1(-1.3331) + 3(1.1760) \\ 4(-1.3331) + 10(1.1760) \end{pmatrix} = \begin{pmatrix} 2.1964 \\ 6.4326 \end{pmatrix}$$

$$\hat{y} = \begin{pmatrix} 2.1964 \\ 6.4326 \end{pmatrix} + \begin{pmatrix} 1.0180 \\ 0.9121 \end{pmatrix} = \begin{pmatrix} 3.2149 \\ 7.3447 \end{pmatrix}$$

2. Error Vector (e) and Cost (J):

$$e = \hat{y} - y = \begin{pmatrix} 3.2149 \\ 7.3447 \end{pmatrix} - \begin{pmatrix} 5 \\ 6 \end{pmatrix} = \begin{pmatrix} -1.7851 \\ 1.3447 \end{pmatrix}$$

$$J = \frac{1}{2}\left[\left[(-1.7851)^2 + (1.3447)^2\right]\right] = \frac{3.1866 + 1.8082}{2} = 2.4974$$

3. Gradients:

$$\frac{\partial J}{\partial m} = X^T e = \begin{pmatrix} 1 & 4 \\ 3 & 10 \end{pmatrix} \begin{pmatrix} -1.7851 \\ 1.3447 \end{pmatrix} = \begin{pmatrix} 1(-1.7851) + 4(1.3447) \\ 3(-1.7851) + 10(1.3447) \end{pmatrix} = \begin{pmatrix} 3.5937 \\ 8.0917 \end{pmatrix}$$

$$\frac{\partial J}{\partial b} = e = \begin{pmatrix} -1.7851 \\ 1.3447 \end{pmatrix}$$

4. Updates:

$$m^{(3)} = m^{(2)} - 0.01 \frac{\partial J}{\partial m} = \begin{pmatrix} -1.3331 \\ 1.1760 \end{pmatrix} - \begin{pmatrix} 0.0359 \\ 0.0809 \end{pmatrix} = \begin{pmatrix} -1.3690 \\ 1.0951 \end{pmatrix}$$

$$b^{(3)} = b^{(2)} - 0.01 \frac{\partial J}{\partial b} = \begin{pmatrix} 1.0180 \\ 0.9121 \end{pmatrix} - \begin{pmatrix} -0.0179 \\ 0.0134 \end{pmatrix} = \begin{pmatrix} 1.0364 \\ 0.8987 \end{pmatrix}$$

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Cost after the third update:

$m^{(3)} = (-1.369037, 1.095583)^T$

$b^{(3)} = (1.036351, 0.898653)^T$.

$Xm^{(3)} = (1(-1.369037) + 3(1.095583),$
$4(-1.369037) + 10(1.095583))^T$
$= (1.917712, 5.479682)^T$

$\hat{y} = Xm^{(3)} + b^{(3)} = (1.917712 + 1.036351, 5.479682$
$+ 0.898653)^T = (2.954063, 6.378335)^T$

$e = \hat{y} - y = (2.954063 - 5, 6.378335 - 6)^T$
$= (-2.045937, 0.378335)^T$

$J = ((-2.045937)^2 + (0.378335)^2)/2$
$= (4.1859 + 0.1431)/2 = \underline{2.1645}$.

Summary

Iteration	m_1	m_2	b_1	b_2	Cost J
0	-1.0000	2.0000	1.0000	1.0000	61.0000
1	-1.4500	0.8700	0.9900	0.8900	6.5033
2	-1.3331	1.1765	1.0185	0.9121	2.4979
3	-1.3690	1.0956	1.0364	0.8987	2.1645

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Trend: Are m and b moving in a direction that reduces the error?

- Yes, the cost falls at every update: 61 -> 6.50 -> 2.50 -> 2.16. Each update moves m and b in a direction that reduces the error.

m_1 moves from -1 to about -1.37, m_2 falls from 2 to about 1.10, and b_1 and b_2 stay close to 1 and 0.9. The changes get smaller each time, which shows the values are settling toward a minimum.

Iteration 1: - Andrew Thon Riem Alier.

Iteration 2: - Kevin Ineza

Iteration 3: - Umutoni Kenia.

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